Hodgkin-Huxley Neuron Model implementation project report

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Project description:

This project implements a single-compartment Hodgkin–Huxley neuron (Na+ activation/inactivation: m, h; K+ activation) and simulates it with forward Euler using a small time step dt. I run current-clamp and voltage-step protocols to study spiking dynamics, the firing rate vs DC current (f–I) and rheobase, the strength–duration curve (Weiss–Lapicque fit for Irh and chronaxie tau_c), validate gating kinetics against analytic steady-state and time-constant curves (x_inf(V), tau_x(V)), and confirm numerical reliability with a dt-sensitivity check.

Limitations:
point-neuron (single compartment, no morphology), no synapses/network context, deterministic channels (no noise/modulation), idealized voltage clamp, and protocol-/analysis-dependent thresholds; kinetics extracted from sparse (V_test) with simple tau estimation and Euler integration. Good at chosen (dt) but not universally robust).
Mathematical frameworks used:

- Nonlinear ODE dynamical system (HH equations for V, m, h, n)
- Numerical integration (forward Euler with step dt)
- Event/threshold detection (spike times, firing rate)
- Bracketing/root-finding (bisection for pulse threshold Ith(d))
- Curve fitting/regression (Weiss–Lapicque linearization for I_rh and tau_c)
- Voltage-clamp system identification/validation (estimating x_inf(V), tau(V))

Experiments:

A compact series of experiments was run on the Hodgkin–Huxley (HH) single-compartment neuron to connect Na^+/K^+ gating kinetics (m, h, n) to spiking behavior and numerical reliability.

1. Current clamp
I injected depolarizing inputs and observed how spikes emerge from the channel dynamics. A sufficiently strong stimulus triggers the HH sequence: fast Na^+ activation (m↑) drives the upstroke, Na^+ inactivation (h↓) shuts off the inward current, and slower K^+ activation (n↑) repolarizes the membrane and sets recovery.

Essential formulas:

Membrane capacitance

$$C_m \frac{dV}{dt} = I_{\text{ext}}(t) - I_{\text{Na}} - I_{\text{K}} - I_{\text{L}}$$

$$I_{\text{Na}} = \bar{g}_{\text{Na}} m^3 h (V - E_{\text{Na}}), \quad I_{\text{K}} = \bar{g}_{\text{K}} n^4 (V - E_{\text{K}}), \quad I_{\text{L}} = g_L (V - E_L)$$

Euler update:

$$V_{k+1} = V_k + \Delta t \frac{dV}{dt}, \quad x_{k+1} = x_k + \Delta t \frac{dx}{dt}$$

2. Firing rate vs DC current (f–I)
I measured steady firing rate as a function of constant injected current and estimated the rheobase. Below threshold the model stays silent; above rheobase it transitions to tonic spiking with firing rate increasing with current, consistent with HH excitability and the role of K^+ activation in setting the inter-spike interval.

Essential formulas:

Spike counts in DC:

$$f(I_0) = \frac{N_{\text{spikes}}}{T_{\text{win}}} \ [\text{Hz}]$$

Rheobase:

$$I_{\text{rh}}^{(f-I)} = \min\{I_0 : \ f(I_0) > 0\}$$

3. Strength–duration curve
I computed the spike threshold current as a function of pulse width and fitted the Weiss–Lapicque relation. The threshold drops rapidly as pulse duration increases and approaches an asymptotic “pulse rheobase” for long pulses, while the fit provides a chronaxie timescale that summarizes temporal integration.

Essential formula:

Weiss-Lapique:

$$I_{\mathrm{th}}(d) = I_{\mathrm{rh}}\left(1 + \frac{\tau_c}{d}\right)$$

4. Voltage-clamp validation of gating kinetics
Using voltage steps, I extracted steady-state gates and time constants and compared them to analytic HH curves. After correcting the fitting window to use only the step interval, the clamp-estimated m∞/h∞/n∞ and τm/τh/τn match the expected voltage dependence, confirming the gating-kinetics implementation and extraction method.

Essential formulas:

Gates evolution under voltages:

$$\frac{dx}{dt} = \alpha_x(V_{\mathrm{cmd}})(1 - x) - \beta_x(V_{\mathrm{cmd}})x$$

Analytical targets:

$$x_\infty(V) = \frac{\alpha_x(V)}{\alpha_x(V) + \beta_x(V)}, \quad \tau_x(V) = \frac{1}{\alpha_x(V) + \beta_x(V)}$$

5. dt sensitivity check
I repeated simulations with multiple time steps and found near-overlapping voltage traces across dt values, indicating numerical convergence. The small differences in voltage error and spike timing confirm the chosen dt is sufficiently small for accurate HH dynamics under these protocols.

Essential formulas:

Errors against the finest step:

$$e(t) = V_{dt}(t) - V_{dt_{\min}}(t)$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{k=1}^{N} e_k^2}, \quad \max|e| = \max_k |e_k|$$

Results of the experiments:

- Baseline sanity check (I_ext=0): the membrane stays near rest (~ -65 mV) with only a small transient, indicating stable dynamics and consistent initialization.
- Gating curves: analytic m_∞, h_∞, n_∞ and τ_m, τ_h, τ_n show the expected HH voltage dependence ($m_\infty\uparrow$ with V, $h_\infty\downarrow$ with V, $n_\infty\uparrow$; τ_m fast, τ_h peaks around ~ -70 mV, τ_n slower and decreases with depolarization).
- Clamp validation: after fixing the steady-state/tau extraction window, clamp-estimated points align closely with analytic gate curves and time constants, confirming the protocol + fitting are working.
- f–I curve (current clamp): clear rheobase at $I_0 \approx 7.0\ \mu A/cm^2$; above threshold firing appears immediately at a finite rate (e.g., ~55.6 Hz at $I_0=7$), then increases with injected current.
- Strength–duration: threshold current decreases strongly with pulse width; Weiss–Lapicque fit yields $I_{rh} \approx 1.202\ \mu A/cm^2$ and chronaxie $\tau_c \approx 5.182$ ms (threshold flattens for long pulses).
- dt sensitivity: traces for dt = 0.02, 0.01, 0.005 ms overlap closely; relative to dt=0.005 ms, dt=0.01 ms gives small error (RMSE ~0.1645 mV, max|err| ~2.61 mV) and very small first-spike timing shift (~0.0083 ms), so dt=0.01 ms is a good accuracy/cost tradeoff.